

Secure File Sharing System Report

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1. Objective

To develop a secure file-sharing portal ensuring end-to-end encrypted file exchanges between Internee.pk and external parties, leveraging modern security protocols.

2. Tools Used

Tool	Purpose
Python Flask	Web portal backend
AES-256-CBC	File encryption
HMAC-SHA256	Signed URL generation
Kaggle Dataset	Test data
cryptography lib	Encryption implementation

3. Architecture

User Upload → AES-256 Encrypt → Encrypted Storage
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Signed URL Generated (15 min expiry)

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User Download → Verify Signature → AES-256 Decrypt → File

4. Encryption Implementation

All files are encrypted using **AES-256-CBC** before storage. The encryption key is derived from a secure password using SHA-256. The IV (Initialization Vector) is randomly generated per file and prepended to the ciphertext, ensuring unique encryption outputs even for identical files.

5. Signed URLs

Temporary signed URLs are generated using **HMAC-SHA256** with a 15-minute expiry. The signature prevents unauthorized access and URL tampering. Requests with expired or tampered URLs are automatically rejected by the system.

6. Project Summary & Verification

- **Project Summary:** The system utilizes AES-256-CBC encryption to provide end-to-end security, ensuring that data is encrypted at rest within the storage repository.
- **Security Mechanism:** Time-limited Signed URLs were implemented to enforce access control, ensuring that file links expire automatically after 15 minutes.
- **Verification:** System integrity was validated by comparing original versus decrypted file content and checking file sizes, confirming zero data loss during the encryption-decryption cycle.

7. Results

Deliverable	Status
Secure file portal developed	Complete
AES-256 encryption on upload	Complete

AES-256 decryption on download	Complete
Signed URLs implemented	Complete
Kaggle test dataset used	Complete
File integrity verified	Complete